

Online Gym Management System Using Spring Boot

A Project Report

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IN

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**Faculty of Engineering and Technology, Institute of Technical Education and
Research**

SIKSHA 'O' ANUSANDHAN (DEEMED TO BE) UNIVERSITY

Bhubaneswar, Odisha, India

(December 2024)



CERTIFICATE

This is to certify that the project report titled “**Online Gym Management System**” being submitted by (**Swoyam Prakash Mohanty** and **21410121**) to the Institute of Technical Education and Research, Siksha ‘O’ Anusandhan (Deemed to be) University, Bhubaneswar for the partial fulfillment for the degree of Bachelor of Technology in Computer Science and Engineering is a record of original confident work carried out by them under my/our supervision and guidance. The project work, in my/our opinion, has reached the requisite standard fulfilling the requirements for the degree of Bachelor of Technology.

The results contained in this project work have not been submitted in part or full to any other University or Institute for the award of any degree or diploma.

(Name and signature of the Project Supervisor)

Department of Computer Science and Engineering

Faculty of Engineering and Technology;
Institute of Technical Education and Research;
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I am also profoundly thankful to my group members for their dedication, teamwork, and relentless efforts, which were vital to the project's success.

Lastly, I am grateful to everyone who directly or indirectly contributed to this project and helped make it a reality.

Place:

Signature of Student

Date:

DECLARATION

We declare that this written submission represents our ideas in our own words and where other's ideas or words have been included, we have adequately cited and referenced the original sources. We also declare that we have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea/fact/source in our submission. We understand that any violation of the above will cause for disciplinary action by the University and can also evoke penal action from the sources which have not been properly cited or from whom proper permission has not been taken when needed.

Signature of Student with Registration Number

Date: _____

REPORT APPROVAL

This project report titled “**Online Gym Management System**“ submitted by **Swoyam Prakash Mohanty** is approved for the degree of *Bachelor of Technology in Computer Science and Engineering*.

Examiner(s)

Supervisor

Project Coordinator

PREFACE

The project, *"Online Gym Management System using Spring Boot,"* aims to design an efficient and user-friendly platform for managing gym operations. The purpose of this study is to gain practical knowledge of developing a full-stack application using modern frameworks and tools while understanding how Spring Boot simplifies the process of application setup, configuration, and deployment. By working on this project, we also explored how such systems streamline gym management tasks like member registration, fee management, and scheduling, ensuring smooth operational workflows. The project methodology incorporates various technologies:

- **Java Full Stack** for efficient backend and frontend integration.
- **Spring Boot** for rapid application development by minimizing infrastructure overhead.
- **MySQL** as the database for secure and structured data management.
- **CSS, JavaScript, and JSP** for a responsive and interactive frontend design.
- **Maven** as a build automation tool, **Eclipse IDE** for code development, and **Tomcat** as the application server.

This study has provided insights into developing scalable web applications and enhanced our understanding of integrating various technologies to create a functional and robust system.

INDIVIDUAL CONTRIBUTIONS

Swoyam Prakash Mohanty	Front-end ,Research and Analysis
Ronik Kumar Das	Front-end , Back-end and connectivity
Saugat Mohanty	Testing, Front-end, Report
Ankit Mishra	Front-end, Back-end, Report

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INTRODUCTION

1. Project Overview / Specifications

The project, *"Online Gym Management System using Spring Boot,"* is designed to streamline gym operations by offering a comprehensive platform for managing memberships and schedules. , the system provides an easy-to-use interface for gym administrators to manage members, classes, and schedules. This application streamlines administrative tasks, improves communication between staff and members, and ensures a smooth operation of the gym's day-to-day activities.

Project Objectives

- To develop an online platform for gym management that simplifies operations such as member registration, fee management, and workout scheduling.
- To enhance the overall efficiency and accuracy of gym management processes.

Key Features

This website provides Features like

- Member Management
- Gym Item Management
- Slot scheduling and management
- Authentication and Authorization
- User Feedback

Modules and Functionalities

Modules :

1. Admin

Key Functionalities:

- Manage gym items
- Manage slots
- Manage booking
- Manage users
- Manage slot reports
- Review user feedback
- View profile

2. Member

Key Functionalities:

- View profile
- View and book slots
- Cancel bookings
- Provide feedback on services and facilities

2. Motivations

In today's fast-paced world, fitness centers and gyms play a crucial role in promoting healthy lifestyles. However, managing gym operations manually can lead to inefficiencies, errors, and difficulties in scaling. This motivated the development of the *"Online Gym Management System using Spring Boot,"* a project aimed at simplifying and automating the management of memberships, payments, and scheduling for gyms.

The primary motivation behind this project is to address the limitations of traditional manual gym management practices, such as error-prone record-keeping and time-consuming processes. With the rapid adoption of technology in various domains, there is a growing demand for digital solutions that enhance operational efficiency.

Additionally, the project serves as a learning opportunity to gain hands-on experience with modern development frameworks like Spring Boot and Java Full Stack. It enables the application of theoretical knowledge to solve real-world problems, fostering a deeper understanding of full-stack web application development.

3. Uniqueness of the Work

This project is unique due to its comprehensive approach to addressing gym management challenges through innovative use of technology. By leveraging the Spring Boot framework, the system offers a streamlined development process, enhanced scalability, and faster deployment.

The integration of an interactive frontend (CSS, JavaScript, JSP) with a secure and reliable backend (Java, MySQL) ensures a smooth user experience. Key features like real-time customizable membership plans, and intuitive workout scheduling distinguish this system from existing solutions.

Furthermore, the adaptability of this system to cater to small and medium-sized gyms, combined with its potential for future enhancements, sets it apart. This uniqueness lies in its focus on practicality, user-centric design, and the ability to handle dynamic requirements, making it a standout solution in the fitness technology domain.

LITERATURE SURVEY

1. Existing System

In the current scenario, gym management is often handled through manual processes or generic management software. These manual methods typically involve maintaining physical registers or spreadsheets for tracking memberships, payments, and workout schedules. While such methods may work for small-scale operations, they are prone to errors, time-consuming, and lack real-time updates.

On the other hand, some existing gym management systems offer digital solutions. These include software with features like member management, payment tracking, and schedule planning. However, these systems often come with high costs, complex configurations, and limited customization options. Furthermore, many of these solutions are designed for large-scale gyms, leaving smaller gyms struggling to find affordable and tailored options.

2. Problem Identification

The existing systems present several challenges, which are summarized as follows:

1. Manual Processes:

- Prone to human error in record-keeping and calculations.
- Time-intensive and inefficient for growing gyms.
- Lack of centralized data access and management.

2. Limitations of Existing Software:

- High subscription costs make them inaccessible for small and medium-sized gyms.
- Limited customization to meet the specific needs of individual gyms.
- Complex user interfaces, requiring extensive training for staff.

3. Scalability and Flexibility Issues:

- Manual systems fail to scale with the growth of gyms.
- Existing software may not adapt well to new requirements or evolving user needs.

These limitations in the existing systems highlight the need for a solution that is affordable, scalable, and tailored for small to medium-sized gyms. The *"Online Gym Management System using Spring Boot"* addresses these challenges by providing an efficient, user-friendly, and customizable platform for managing gym operations.

MATERIALS AND METHODS

1. Description

The *Online Gym Management System* is a web-based platform developed to automate gym operations and enhance efficiency. The primary goal of the system is to manage memberships, workout schedules, and other administrative tasks seamlessly. By leveraging Spring Boot and Java Full Stack technologies, the project aims to deliver a high-performance application with a responsive and intuitive user interface.

The system is designed to address the challenges of manual gym management by offering:

- Simplified member registration and login processes.
- Centralized management of memberships .
- Role-based access control for administrators, trainers, and gym members.
- A personalized dashboard for members to view workout plans, schedules, and Bookings.

This section outlines the resources, technologies, tools, and methodologies employed in the development of the *Online Gym Management System using Spring Boot*. This system was designed to streamline gym operations by integrating modern software tools and frameworks to achieve a robust, scalable, and user-friendly platform. The purpose of this section is to provide a comprehensive understanding of the processes and approaches adopted to ensure the successful implementation of the project.

2. Schematic Layout/Model Diagram

The system architecture is structured into three main layers:

1. Frontend Layer:

- Built using JSP, CSS, and JavaScript to create an engaging and user-friendly interface.
- Provides the interface for gym members and staff to interact with the system.

2. Backend Layer:

- Developed using Spring Boot and Java to handle business logic and process user requests.
- Ensures secure communication between the user interface and the database.

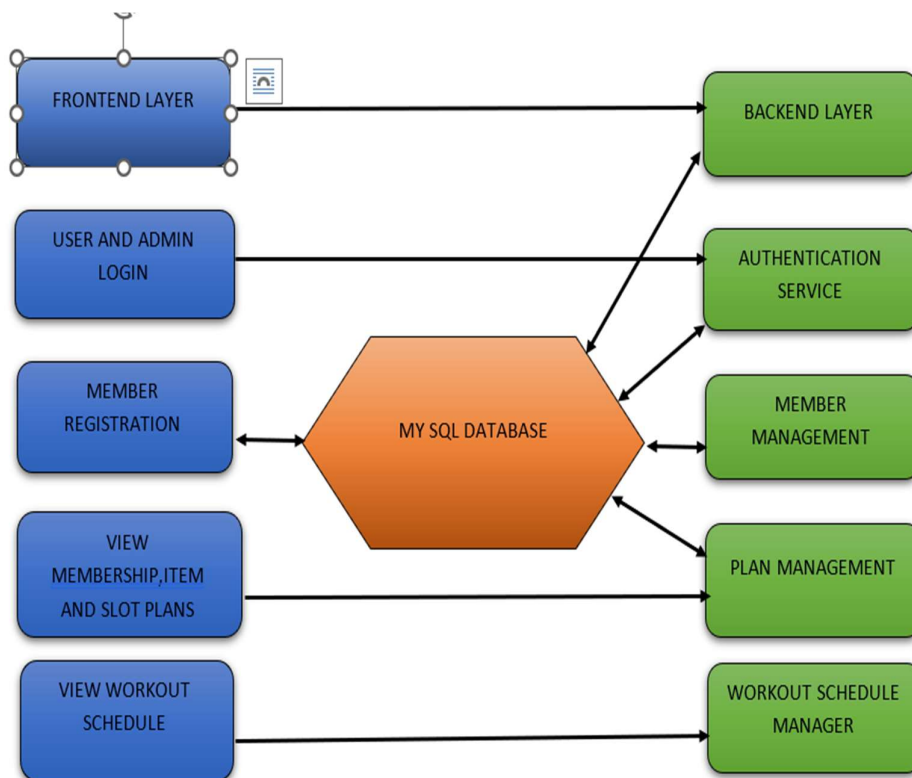


Fig 1

3. Database Layer:

- MySQL is used to store and manage all gym-related data, including member details, payment records, and workout schedules.
- Data is organized and retrieved efficiently to support real-time updates.

Some Databases :

- Feedback table

Field Name	Data Type	Relational Key
f_id	bigint	Primary Key
User_name	varchar (255)	Not Null
content	longtext	Not Null

Table 1

- Gym Items

Field Name	Data Type	Relational Key
item_id	bigint	Primary Key
item_name	varchar (255)	Not Null
total_seat	int	Not Null

Table 2

- Gym Users and Admins

Field Name	Data Type	Relational Key
username	varchar (255)	Primary Key
email	varchar (255)	Not Null
first_name	varchar (255)	Not Null
last_name	varchar (255)	Not Null
password	varchar (255)	Not Null
type	varchar (255)	Not Null

Table 3

- Gym Slot

Field Name	Data Type	Relational Key
slot_id	bigint	Primary Key
pricing	double	Not Null
slot_time	varchar (255)	Not Null

Table 4

- Slot Items

Field Name	Data Type	Relational Key
item_id	bigint	Primary Key
slot_id	bigint	Primary Key
seat_booked	int	Not Null

Table 5

4. ER Diagram

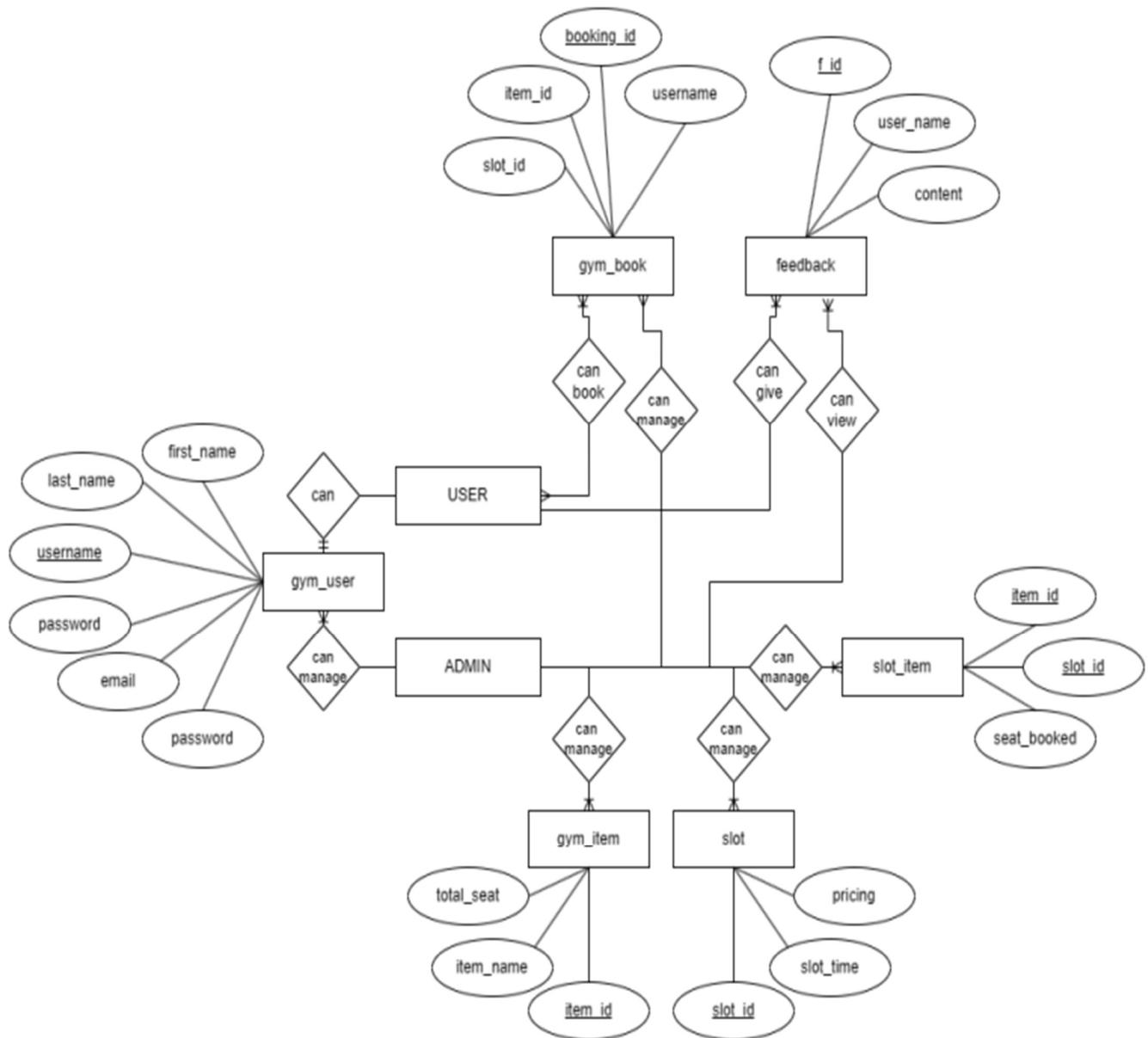


Fig 2

3. Methods Used

The project follows an iterative development methodology to ensure flexibility and continuous improvement. The steps involved are:

1. Requirement Analysis:

- Understanding the challenges of manual gym management and identifying key requirements for automation.
- Collecting inputs from gym owners and staff to design a tailored solution.

2. System Design:

- Creating wireframes and architectural diagrams to visualize the system's structure.
- Designing the database schema to support efficient data storage and retrieval.

3. Implementation:

- Developing the backend using Spring Boot to handle business logic and integrate with the database.
- Creating the frontend using JSP, CSS, and JavaScript for a responsive and interactive user experience.

4. Testing:

- Conducting unit testing to validate individual components.
- Performing integration testing to ensure smooth communication between the frontend, backend, and database.

5. Deployment:

- Deploying the application using Apache Tomcat server for hosting and making it accessible to users.

4. Tools Used

The development of the *Online Gym Management System* utilized the following tools and technologies:

- **Spring Boot:** A framework for building robust and scalable backend applications.

❖ Spring Boot Features Used here :

1. **Spring Boot Starter Dependencies** - Spring Boot Starter Web: Used for building web applications with embedded servers like Tomcat, eliminating the need for explicit server setup. - Spring Boot Starter Data JPA: Used for working with relational databases like MySQL in a simple and efficient way.
2. **Embedded Server** - The project utilizes Apache Tomcat as an embedded server, making it easier to run the application without deploying it externally
3. **Spring MVC** - Implements the Model-View-Controller (MVC) architecture for separating business logic, presentation, and data access layers.
4. **Spring Data JPA** - Used for database operations such as CRUD (Create, Read, Update, Delete) without writing boilerplate SQL queries. - The integration with MySQL ensures efficient data management and processing.
5. **Dependency Management** - Maven is used as a build tool, and Spring Boot simplifies dependency management by automating

6. **Spring Security** - Adds security to the application by implementing role-based access control (e.g., for admin and users). - Handles login authentication and authorization for different users.
7. **ApplicationConfiguration** Spring Boot's application.properties or application.yml file is used for centralizing and customizing configurations such as database connections, server ports, etc.
8. **RESTful APIs** - Spring Boot provides features for creating and managing REST APIs, allowing seamless communication between frontend and backend.
9. **Exception Handling** - Custom exception handling is supported using Spring Boot's @ExceptionHandler annotations.
10. **Automatic Configuration** - Spring Boot automatically configures components like Hibernate, JPA, and data sources, reducing the amount of manual configuration.

- **Java Full Stack:** The primary technology stack used for both backend and frontend development.
- **MySQL:** A relational database for secure and efficient data management.
- **JSP, CSS, and JavaScript:** For designing an interactive and responsive user interface.
- **Maven:** A build automation tool for managing project dependencies and builds.
- **Eclipse IDE:** The development environment used for writing and debugging code.
- **Apache Tomcat:** A server for deploying and running the web application.

5. Working Process Description

The *Online Gym Management System* operates as follows:

1. User Registration and Login:

- Users (gym members, staff, and administrators) can register by providing necessary details.
- Secure login credentials are generated, and role-based access is assigned.

2. Membership, Items, Prices and Slot Management:

- Administrators can create, update, and manage different plans.
- Members can view and select plans based on their preferences.

3. Booking Tracking:

- The system records bookings made by members and updates the database in real-time.
- Administrators can generate reports, while members can view their booking and slot history.

4. Workout Scheduling:

- Trainers can create personalized workout schedules for members.
- Members can view their assigned schedules and track their progress.

5. Data Management:

- All user data, membership details, and payment records are securely stored in the MySQL database.
- The backend, powered by Spring Boot, handles data processing and ensures real-time updates.

This section highlights the systematic approach and tools used to develop and implement the *Online Gym Management System*. It reflects the use of modern technologies and efficient processes to deliver a functional and user-centric application.

RESULTS AND OUTPUTS

1. System Specification

The *Online Gym Management System* has been developed to meet the following specifications:

- **System Type:** Web-based application.
- **Frontend Technologies:** JSP, CSS, and JavaScript for an interactive and responsive interface.
- **Backend Technologies:**
 - **Framework:** Spring Boot for efficient backend development.
[Java Servlet Version 4.0.1]
 - **Programming Language:** Java.
[JDK Version 17]
- **Database:** MySQL for secure and structured data storage.
[MySQL Version 8]
- **Build Tool:** Maven for dependency management and project builds.
[Maven Version 2.7.16]
- **Development Environment:** Eclipse IDE for writing, debugging, and testing code.
[Eclipse IDE for Enterprise Java and Web Developers - 2024-03]
- **Server:** Apache Tomcat for hosting and running the application.
[Tomcat Version 9.0.65]

2. Experimental Outcomes

The system has been developed, tested, and validated to achieve the following outcomes:

User Management

- **Registration and Login:** Members and gym staff can successfully register and log in to the system.
- **Role-based Access Control:** Implemented to manage staff and member access, ensuring secure data handling.
- **User Dashboard:** Features an intuitive and responsive UI for seamless user interaction.

Membership Plan Handling

- Gym staff can create, manage, and assign membership plans to members.
- Members can view available plans, make selections, and make payments online.

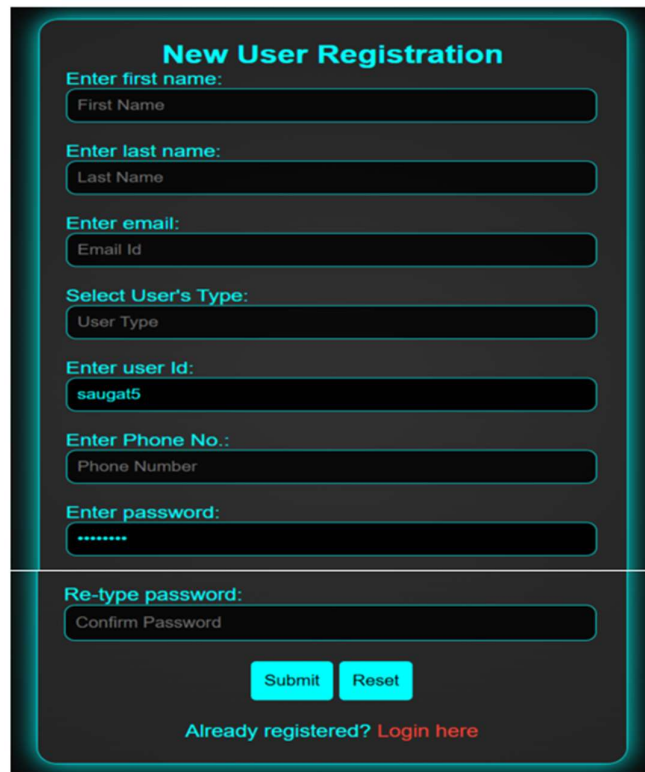
Workout Scheduling

- Gym staff can assign workout schedules to members, which are visible on their dashboard.
- Members can view their personalized workout schedules and interact with trainers through the system.

System Performance

- The system has been tested for performance, handling multiple concurrent users efficiently.
- Backend logic and data processing are optimized for minimal delays and optimal user experience.

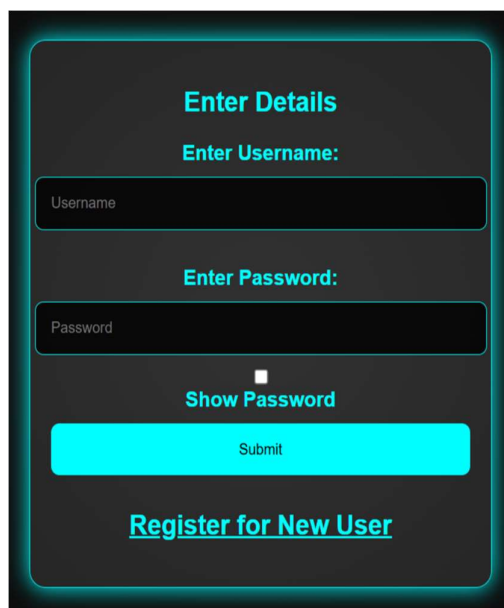
3. UI of Website



The image shows a 'New User Registration' form with a dark background and a glowing cyan border. The form contains the following fields and elements:

- New User Registration** (Title)
- Enter first name:** Input field with placeholder 'First Name'.
- Enter last name:** Input field with placeholder 'Last Name'.
- Enter email:** Input field with placeholder 'Email Id'.
- Select User's Type:** Input field with placeholder 'User Type'.
- Enter user Id:** Input field with placeholder 'saugat5'.
- Enter Phone No.:** Input field with placeholder 'Phone Number'.
- Enter password:** Input field with placeholder '*****'.
- Re-type password:** Input field with placeholder 'Confirm Password'.
- Submit** and **Reset** buttons.
- Already registered? [Login here](#)** (Link).

Fig 3. New User Registration



The image shows a login form with a dark background and a glowing cyan border. The form contains the following fields and elements:

- Enter Details** (Title)
- Enter Username:** Input field with placeholder 'Username'.
- Enter Password:** Input field with placeholder 'Password'.
- Show Password** (Text with a small square icon).
- Submit** button.
- [Register for New User](#)** (Link).

Fig 4. Login Pag

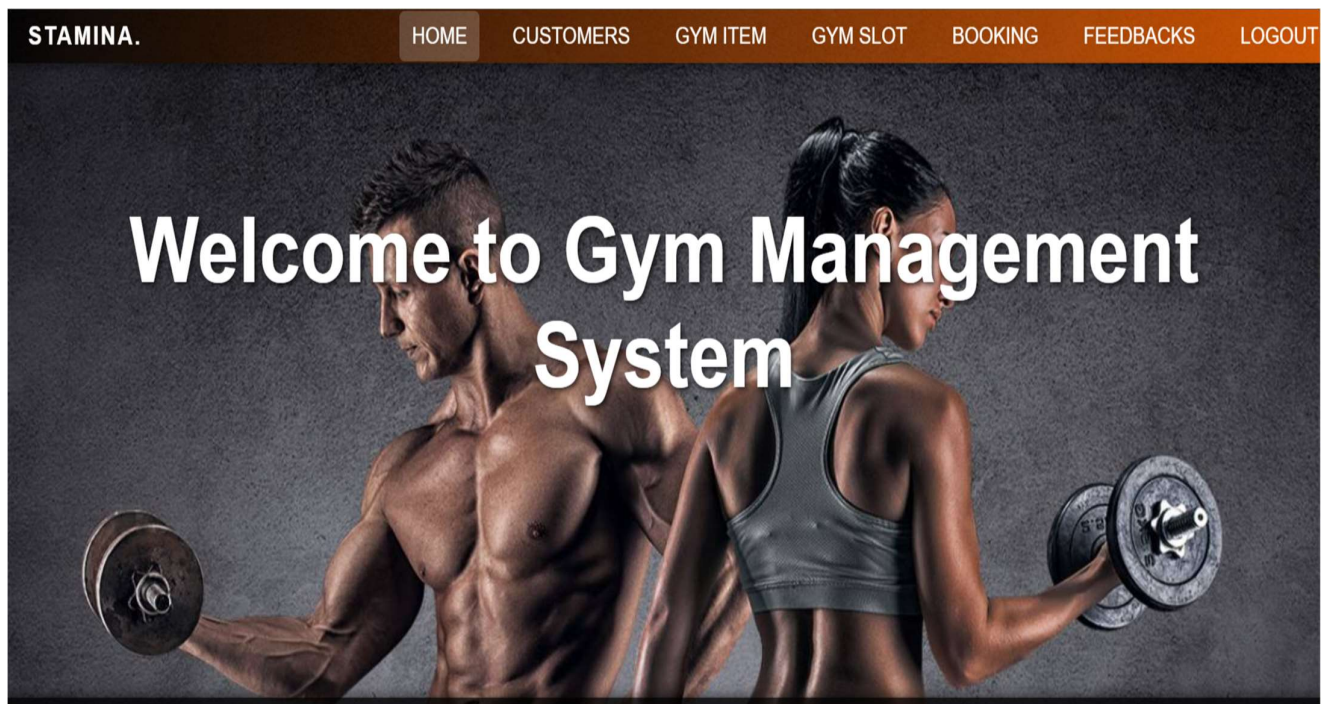


Fig 5. Home Page

ADMIN SIDE UI

Manage Exercises

Exercise Name:

Total Seat:

Add Exercise

Fig 6. Item Details

{Admin can enter gym item name and Total seats}

Edit Exercise

Exercise Name: Swimming

Swimming

Total Seat:

30

Update Exercise

Fig 7. Edit Item Details
{Admin can edit gym item details }

GYM ITEMS				
Item Number	Item Name	Total Seat/Slot	Edit Seat	Add To Slots
101	Swimming	30	Edit	Add To Slots
102	Pushups	2	Edit	Add To Slots
103	Running	5	Edit	Add To Slots
104	Cardio	1	Edit	Add To Slots
105	Bench Press	4	Edit	Add To Slots
Return				

Fig 8. View Item Details
Admin can view item details

Slots

Slot Id	Slot Time	Slot Price	Edit
1	6am	800.0	Edit
2	7am	1000.0	Edit
3	8am	1200.0	Edit
4	9am	1500.0	Edit
5	10am	2000.0	Edit

[RETURN](#)

Fig 9. Time Slot and Price Management

Bookings

Booking Id	Slot ID	Item ID	Username	Select
1000001	1	104	john	☐
1000002	1	102	rose	☐
1000003	2	104	john	☐

[CANCEL](#)
[RETURN](#)
[RESET](#)

Fig 10. View Booked Slots

Book Slot

Slot Id	Slot Time	Price
1	6am	800.0

Select User Id:

Item No	Item Name	Total Slot	Seat Vacant	Select
101	Swimming	30	30	☐
102	Pushups	2	1	☐
103	Running	5	5	☐
104	Cardio	1	0	☐
105	Bench Press	4	4	☐

BOOK

RETURN

RESET

Fig 11. View Items Selected along Slots

Customer Side UI

Slots

Slot Id	Slot Time	Slot Price	Enquire
1	6am	800.0	Enquire
2	7am	1000.0	Enquire
3	8am	1200.0	Enquire
4	9am	1500.0	Enquire
5	10am	2000.0	Enquire

RETURN

Fig 12. View Slot Availability and Prices

Book Slot				
Slot Id		Slot Time		Price
1		6am		800.0
Item No	Item Name	Total Slot	Seat Vacant	Select
101	Swimming	30	30	☐
102	Pushups	2	1	☐
103	Running	5	5	☐
104	Cardio	1	0	☐
105	Bench Press	4	4	☐

Fig 13. Book Slots



Fig 14. Check Booking Successfull or Failed

CONCLUSION

The development of the **Online Gym Management System using Spring Boot** has been a comprehensive learning experience, integrating theoretical knowledge with practical implementation. This project has successfully demonstrated the ability to streamline gym operations by providing an efficient, user-friendly platform for managing members, schedules, payments, and workout plans. By leveraging modern technologies such as Spring Boot for the backend, MySQL for database management, and Java full-stack development for the application, the system ensures seamless performance and scalability.

The system addresses the challenges of traditional gym management systems, including manual record-keeping, inefficiencies in communication, and difficulty in managing schedules and payments. By introducing automated functionalities and an interactive user interface, it enhances the overall user experience for both administrators and gym members.

This project has not only highlighted the importance of structured software development methodologies but has also emphasized the significance of collaboration, problem-solving, and adaptability in addressing real-world challenges. With further improvements and additional features, the system has the potential to be deployed in real-world scenarios, offering a robust solution for gym management.

In conclusion, the gym management system using Spring Boot is a comprehensive solution that addresses the key challenges faced by gym administrators and members. It enhances operational efficiency, improves member engagement, and provides a solid foundation for future growth and innovation.

APPENDICES

Appendix 1: Controllers

UserController.java

Handles user-related operations.

AdminController.java

Handles administrative operations.

GymController.java

Manages gym-specific functionalities like scheduling and booking.

Appendix 2: Models

User.java

Defines the `User` entity, representing users in the system.

Membership.java

Defines membership plans and related attributes.

Booking.java

Represents user bookings.

Appendix 3: Configuration Files

application.properties

Contains the Spring Boot application's configuration settings, including database configurations and port settings.

log4j.properties

Specifies logging configurations for the application.

Appendix 4: Views (Frontend Code)

login.jsp

Provides the user interface for the login page.

dashboard.jsp

Displays the dashboard for both users and administrators.

membership.jsp

Allows users to view and purchase memberships.

schedule.jsp

Displays available gym schedules for bookings.

Appendix 5: Repositories

UserRepository.java

Handles database operations related to the `User` entity.

MembershipRepository.java

Handles database operations for membership data.

BookingRepository.java

Manages database operations related to booking records.

Appendix 6: Services

UserService.java

Contains business logic for user-related operations.

MembershipService.java

Manages business logic for membership functionalities.

BookingService.java

Implements booking-related business logic.

Appendix 7: Working Process Description

Controller-Service-Repository Interaction

Explains the interaction flow between the three primary layers of the application.

Controllers handle requests, Services process business logic, and Repositories interact with the database.

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REFLECTION OF THE TEAM MEMBERS ON THE PROJECT

1. Write what you learned as a team.

- As a team, we learned to collaborate effectively while developing a robust and user-friendly Online Gym Management System. We gained practical experience in using Spring Boot for backend development, MySQL for database management, and JSP for frontend design. Additionally, we enhanced our understanding of project planning, problem-solving, and implementing a structured MVC architecture, which improved our technical and teamwork skills significantly.

2. Write what you learned as a member.

- As a member, I learned how to build and manage different parts of the project, like coding the backend, handling the database, and designing user interfaces. I also improved my problem-solving skills by fixing issues during development. Working in a team taught me how to share ideas, help others, and stay organized to complete tasks on time.

3. Write a thoughtful paragraph on strengths and weaknesses of your design process.

- The design of the Online Gym Management System using Spring Boot benefited from modern technologies like Java Full Stack, Spring Boot, and MySQL, ensuring efficient development and data management. The clear separation between frontend and backend layers facilitated scalability and maintenance. However, the integration of different technologies posed challenges, requiring careful synchronization. Additionally, the lack of real-time synchronization for workouts and payments could impact user experience, which could be improved in future updates.

THANK YOU !